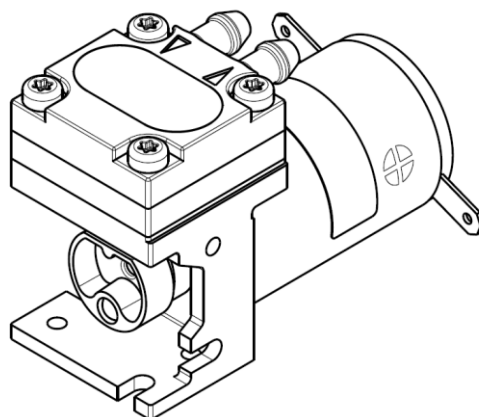


Operating and Installation Instructions

Micro Diaphragm Gas Pumps



Typen range:

NMP 05 S	NMP 09 S	NMP 015 S
NMP 05 M	NMP 09 M	NMP 015 M
NMP 05 L	NMP 09 L	NMP 015 L
NMP 05 B	NMP 09 B	NMP 015 B

You have selected a high-quality KNF product; the following tips will help you operate it safely, and reliably over a long period of time. Carefully study the Operating and Installation Instructions before using the pumps and observe at all times the relevant instructions to avoid dangerous situations. The manual was produced for the serial pumps stated above. With customer-specified projects (pump types starting with "PJM" or "PMM") there could be differences in detail. For customer-specified projects please therefore take into account any agreed technical specifications, as well as these instructions.

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1. Description, Operating Conditions

KNF pumps in the NMP range transfer, evacuate and compress 100% oil-free. In operation they are maintenance-free

See the type- plate or data sheet for full electrical data.

1.1. Operating Conditions

Handling air, gases, and vapours at temperatures between + 5 °C...+ 40 °C.

For maximum permissible operating pressure, ultimate vacuum, and flow capacity see data sheet.

Ambient temperature during operation: between + 5 °C...+ 40 °C

The pumps must not be used in areas where there is a danger of explosion.

Before pumping a medium, the compatibility of materials of pump head, diaphragm and valves with the medium must be checked (for pump materials: see data sheet).

KNF pumps in the NMP range must not be used for liquids. You will find suitable liquid pumps in our Product Program.

If your potential application lies outside the above limits discuss it with our technical adviser (see last page for contact telephone number).

2. Safety

i Note the safety precautions in chapters 3. *Installation and connection*, and 4. *Operation*.

The pumps are built according to the generally recognized rules of technology and in accordance with the occupational safety and accident prevention regulations. Nevertheless, dangers can result during their use which lead to injuries to the user or others, or to damage to the pump or other property.

Only use the pumps when they are in a good technical and proper working order, in accordance with their intended use, observing the safety advice within the Operating and Installation Instructions, at all times.

Personnel Make sure that only trained and instructed personnel or specially trained personnel work on the pumps. This especially applies to assembly, connection and servicing work.

Make sure that the personnel has read and understood the Operating and Installation Instructions, and in particular the "Safety" chapter.

Working in a safety-conscious manner Observe the accident prevention and safety regulations when performing any work on the pump and during operation.

Handling dangerous media When transferring dangerous media, observe the safety regulations when handling these media.

Handling combustible media Be aware that the pumps are not designed to be explosion-proof.

	Make sure the temperature of the medium is always sufficiently below the ignition temperature of the medium, to avoid ignition or explosion. This also applies for unusual operational situations. Note that the temperature of the medium increases when the pump compresses the medium. Hence, make sure the temperature of the medium is sufficiently below the ignition temperature of the medium, even when it is compressed to the maximum permissible operating pressure of the pump. If necessary, consider any external sources of energy, such as radiation, that may add heat to the medium. In case of doubt, consult the KNF customer service.
Environmental protection	Store all replacement parts in a protected manner and dispose of them properly in accordance with the applicable environmental protection regulations. Observe the respective national and international regulations. This especially applies to parts contaminated with toxic substances.
Disposal	Dispose of all packaging in an environmentally-appropriate manner. The packaging materials are recyclable. Dispose of end-of-life equipment in an environmentally friendly manner. Use appropriate waste collection systems for the disposal of end-of-life equipment. Used pumps contain valuable recyclable materials.
EC Directives / Standards	For the purposes of the Machinery Directive 2006/42/EC, pumps are "partly completed machinery," and are therefore to be regarded as not ready for use. Partly completed machinery may not be commissioned until such time as it has been determined that the machine in which the partly completed machinery is to be assembled is in conformity with the provisions of the Machinery Directive 2006/42/EC. The following essential requirements of Annex I of Directive 2006/42/EC (general principles) are applied and observed: - General Principles No. 1 - No. 1.1.2. / 1.1.3. / 1.3.1. / 1.3.3. / 1.3.4. / 1.4.1. / 1.5.8. / 1.5.9. / 1.7.4. / 1.7.4.1. / 1.7.4.3. As these partly completed machinery are OEM-models the power supplies and the equipment for disconnecting and switching-off the partly completed machinery respectively have to be considered when mounting as well as over-current and overload protective gear. In addition a protection against mechanical parts in motion and hot parts, if existing, has to be provided when mounting. The pumps conform to the Directive 2011/65/EU (RoHS2). The pumps conform to the EC Directive 2014/30/EU concerning Electromagnetic Compatibility.



The declaration of the SJ-T 11364-2024 standard „Hazardous Substance Disclosure Table 有害物质含量表“ is carried out in accordance with the specifications detailed in the appendix of this document.

The following harmonized standards have been used:

NMP 05 M NMP 09 M	NMP 05 L NMP 09 L NMP015 L	NMP 05 B NMP 09 B	NMP 015 M
DIN EN 61000-6-3	DIN EN 55022 (Class B)	DIN EN 61000-6-1/2/3/4	DIN EN 55014-1
	DIN EN 61000-4-2/3/4/5/6/8		

Tab. 1

Customer service and repairs	Only have repairs to the pumps carried out by the KNF Customer Service responsible.
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3. Installation

Pumps in the NMP range are OEM models intended for installation in equipment. When installing them make certain that accident prevention regulations, and safety instructions, including those for subsequent operation are observed.

The dimensions of the mountings are given in Data Sheet.

Fit the pump at the highest point in the system, so that condensate cannot collect in the head of the pump - that prolongs working life.

Compare the supply data with the motor data.

With dc motors the wires must be connected to the correct poles.

For brushless DC motors (type designation ending with B):

- Caution; incorrect lead connection will damage motor electronics.
- Correct connection:
 - + red cable
 - black cable

In the electrical installation, arrangements (complying with EN 60335-1) must be made for disconnecting the pump motor from the electrical supply.

EMC compatible installation (for NMP05/09 B)

The maximum length of the connection leads is limited irrespective of whether the device is to be used in an industrial environment or is for domestic use¹⁾.

- All connection leads may not exceed a length of 3 m.

Compliance with the performance defined by the CE mark with respect to emission and immunity requires additional EMC measures:

- Ensuring the allowable emission in domestic use can require the use of an EMC filter and / or an EMC suppressor circuit.
- Ensuring the necessary immunity in industrial uses can require use of an EMC suppressor circuit.

Intended environment	Interference type	Action
Domestic use ¹⁾	Emission	EMC filter
	Immunity	EMC suppressor
Industrial use	Emission	
	Immunity	EMC suppressor

Tab. 2

This table shows which additional EMC measures have to be taken to ensure the equipment fulfils the performance regarding emission and immunity defined by the Directive in the environment in the intended environment.

¹⁾ Domestic means use in homes, business and commercial areas as well as in small businesses.

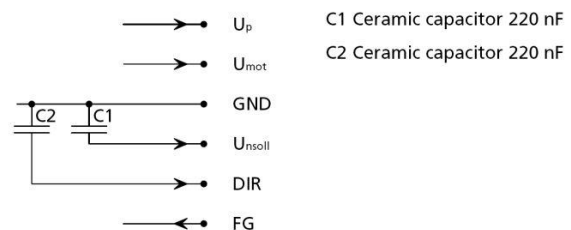
The EMC filter

Circuit diagram 1

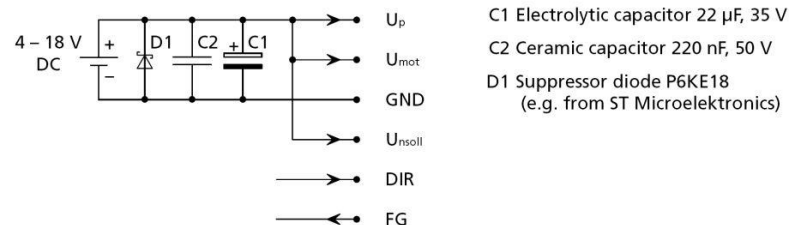


The EMC suppressor circuit

Circuit diagram 2



Circuit diagram 3



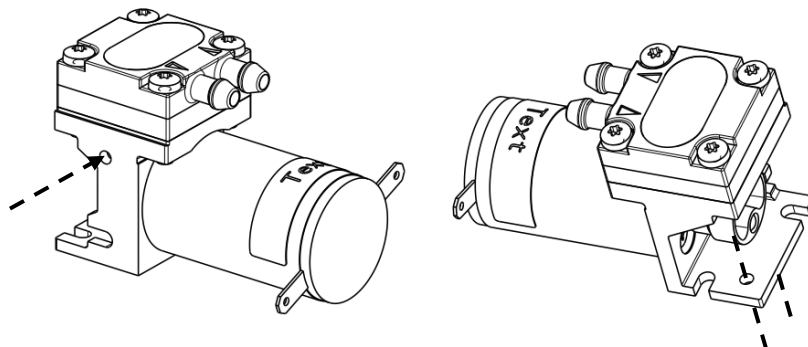
It may not be necessary to implement the additional EMC measures named. If the motor is fed e.g. from a CE-conforming power supply unit which firstly is connected with the motor by the shortest possible connection lead and secondly no other devices are supplied, then the EMC filter (circuit diagram 1) or EMC suppressor circuit (circuit diagram 3) can be dispensable. In this case the power supply unit takes on the function of the EMC filter or the EMC suppressor circuit according to circuit diagram 3.

The same applies accordingly to the EMC suppressor circuit according to circuit diagram 2: The EMC suppressor circuit can be dispensable if other design measures prevent emissions from interfering with the control voltage for the desired speed and the switch input for the direction of rotation.

3.1. Fixation

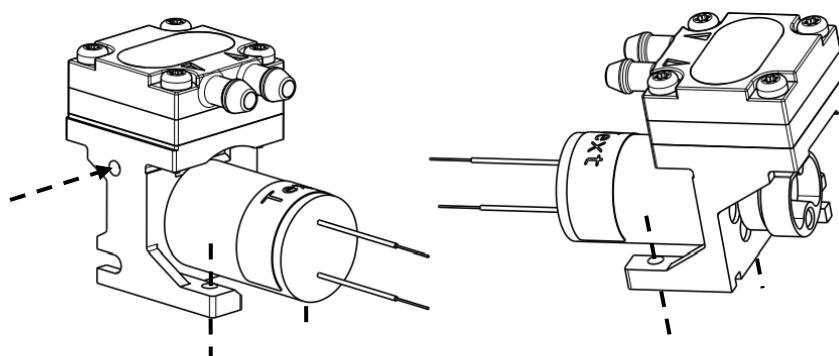
Fixation

The pump has two mounting points, which allow an easy fixation of the pump with 2x Delta PT20 screws.



Fixation (for NMP05 M)

The pump has two mounting points, which allow an easy fixation of the pump with 2x Delta PT20 screws.



4. Operation

Specific safety instructions for the media being handled must be observed.

Before pumping a medium, the compatibility of materials of pump head, diaphragm and valves with the medium must be checked (for pump materials: see data sheet)

The pump must not start against pressure or vacuum. When it is switched on the pressure in the suction and pressure lines must be atmospheric.

The maximum permissible operating pressure (see data sheet) must not be exceeded.

Exception: If the data sheet includes values for intermittent operation, they may be employed briefly.

To prevent the maximum permissible operating pressure being exceeded, restriction or control of the air or gas flow should only be carried out in the suction line.

If restriction or control of the air or gas flow is made on the pressure side ensure that the maximum permissible operating pressure is not exceeded.

When the pump is at a standstill the inlet and exhaust must be at normal atmospheric pressure.

5. Trouble Shooting

Flow, pressure or vacuum are too low, or: Pump produces no flow:

- Connections or lines are blocked.
- There is a leak at a connector, in a line, or in the pump head.
- An external valve is closed, or a filter blocked.
- Liquid (condensate) has collected in the pump head.
 - ▶ Let the pump run for a few minutes pumping air (if necessary for safety reasons: pumping an inert gas.)
 - ▶ Install the pump at the highest point in the system.
- The cross-section of pneumatic lines, or connected components is too small, or they are restricted.
- There is pressure on the pressure side, and at the same time vacuum, or a pressure above atmospheric, on the suction side:
 - ▶ The pump is not designed for this condition

If the pump does not operate properly and you cannot find any of the above faults, send it to the KNF Service Department.

In order for KNF to repair the pump, the customer must provide a statement on the media which were pumped and on pump cleaning. Please fill out the corresponding KNF form and submit it together with the pump.

6. Returns

Preparing for return

1. Flush the pump with air for a few minutes (if necessary for safety reasons: with inert gas) to free the pump head of dangerous or aggressive gases (see Chapter Flushing the pump).



Please contact your KNF sales partner if the pump cannot be flushed due to damages

2. Remove the pump.
3. Clean the pump (see the Chapter Cleaning the pump).
4. Send the pump together with completed Health and Safety Clearance and Decontamination Form to KNF, stating the nature of the transferred medium.
5. Package the device securely to prevent further damage to the product. If necessary, request an original packaging at a charge.

Returns

KNF shall undertake to repair the pump only under the condition that the customer presents a certificate regarding the medium that is pumped and the cleaning of the pump. Please follow the instructions at [knf.com/repairs](https://www.knf.com/repairs).

Contact your KNF sales partner directly if you require additional support for your return service.

7. Appendix

7.1. RoHS Decleration for China

SJ-T 11364-2024 Hazardous Substance Disclosure Table / SJ-T 11364-2024 有害物质含量表

This declaration is valid for the KNF products produced by:
本声明适用于由以下KNF公司生产的产品：

KNF Neuberger GmbHDE-79112 FreiburgGermany

The following information has been made available to comply with SJ-T 11364-2024 the Marking for Control of Pollution Caused by Electronic Information Products as required by China's Management Methods for the Control of Pollution from Electronic Information Products.
提供以下信息旨在遵守中国《电子信息产品污染控制管理办法》所要求的SJ-T 11364-2024《电子信息产品污染控制标识》标准。

Part Name 部件名称	Hazardous Substances Table 有害物质含量表						EFUP / 环境友好使用期限	China RoHS comment / 中国RoHS注释	EU RoHS compliant / 符合欧盟RoHS指令
	Lead 铅 (Pb)	Mercury 汞 (Hg)	Cadmium 镉 (Cd)	Hexavalent Chromium 六价铬 (Cr(VI))	Polybrominated biphenyls 多溴联苯 (PBB)	Polybrominated diphenyl ethers 多溴二苯醚 (PBDE)			
Pump head / 泵头	X	O	O	O	O	O	20	Lead within EU RoHS exemption limit / 铅含量在欧盟RoHS豁免限值以内	Yes / 是
Pump housing / 泵体	X	O	O	O	O	O	20	Lead within EU RoHS exemption limit / 铅含量在欧盟RoHS豁免限值以内	Yes / 是
Pump motor / 泵的电机	X	O	O	O	O	O	20	Lead within EU RoHS exemption limit / 铅含量在欧盟RoHS豁免限值以内	Yes / 是
Additional components for systems / 系统附加部件	X	O	O	O	O	O	20	Lead within EU RoHS exemption limit / 铅含量在欧盟RoHS豁免限值以内	Yes / 是
O: Indicates that this toxic or hazardous substance contained in all of the homogeneous materials for this part is below the limit requirement in SJ-T 11364-2024. 表示该有害物质在该部件所有均质材料中的含量均在 SJ-T 11364-2024 规定的限量要求以下。									
X: Indicates that this toxic or hazardous substance contained in at least one of the homogeneous materials used for this part is above the limit requirement in SJ-T 11364-2024. 表示该有害物质至少在该部件的某一均质材料中的含量超出 SJ-T 11364-2024 规定的限量要求。									
EFUP /环境友好使用期限: Environment Friendly Using Period / 环境友好使用期									

KNF weltweit

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